AA

Abdullah Alzahrani

Software Engineering @ KFUPM

Portfolio

Web Application

Assignment 4 — Final Submission | SWE Foundation

🌐 https://abdullah-zh.github.io/s202265440-Abdullah-Alzahrani-Assignment4/

ID: 202265440

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Project Overview

What It Is

A complete personal portfolio website built with vanilla HTML, CSS, and JavaScript — no frameworks, no dependencies.

Why I Built It

To showcase my engineering skills, leadership experience, and projects in a professional format accessible to anyone online.

Key Goal

Deliver a production-ready, live-deployed web application that looks and works like a real product — not a student project.

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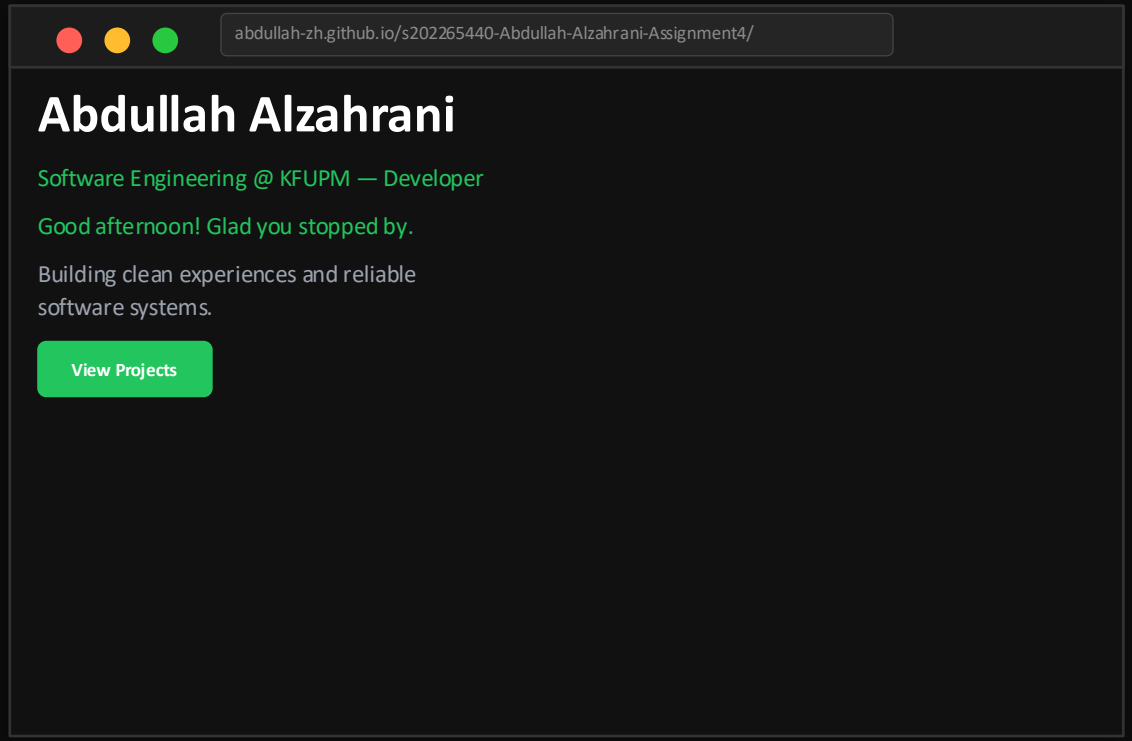
Hero & Navigation



Typewriter Effect

Role cycles: Developer → Designer → Problem Solver

Color Theme Switcher

4 accent colors saved to localStorage

Dark/Light Mode

System-aware theme with persistent preference

Visitor Greeting

Personalized name saved across sessions

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About & Animated Skills

Who I Am

Software Engineering student at KFUPM, Dhahran. Focused on building practical software — from responsive web interfaces to desktop applications.

Leadership: VP at KFUPM Year Book, Content Team Lead at TechHub, and Volunteer Leader for the Shukran 3 Initiative.

Skills

HTML & CSS	92%
JavaScript	78%
Java & JavaFX	80%
Figma & UX Design	85%
Git & GitHub	75%
AI-Assisted Dev	88%

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Interactive Journey Timeline

2022 — Present
Software Engineering Student
KFUPM, Dhahran. Concentration in CX Data Science & Analytics. Expected graduation 2027.

2024 — Present
VP — KFUPM Year Book
Leading the editorial committee, overseeing content production and team coordination.

2024 — Present
Content Team Lead — TechHub
Managing content strategy and technical writing for KFUPM's technology hub.

2024
Volunteer Leader — Shukran 3
Led a team for the Shukran 3 community initiative, coordinating logistics and outreach.

2024
Founders Track Applicant — DTV
Applied to the competitive Founders Track at Dhahran Techno Valley.

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Featured Projects

All | Figma | Java | Design Patterns | UML

Sort by: Default ▾

[Preview]

Table-I

Figma mobile app prototype for restaurant booking with social features.

Figma UML

[Preview]

Student Schedule Visualizer

JavaFX desktop app importing Excel schedules and visualizing class routes.

Java JavaFX

[Preview]

Folder Structure Visualizer

Desktop tool using Composite and Strategy design patterns to render folder hierarchies.

Java Composite

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GitHub API Integration

```
GET  https://api.github.com/users/Abdullah-Zh/repos?sort=updated&per_page=6
```

s202265440-Abdullah-Alzahrani-Assignment4

No description provided.

CSS ★ 0

202265440-AbdullahAlzahrani-Assignment3

No description provided.

CSS ★ 0

Portfolio-Assignment

No description provided.

CSS ★ 0

How It Works

1. Page loads → fetch GitHub API
2. Forks filtered out automatically
3. Cards rendered with name, lang & stars
4. On failure → static fallback cards shown
5. Retry button allows re-fetch without reload

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Technical Architecture

HTML5

Semantic structure & ARIA accessibility

CSS3

Custom properties, Grid, Flexbox, animations

JavaScript ES6+

No frameworks — vanilla JS only

GitHub REST API

Live repository data with fallback

IntersectionObserver

Scroll-triggered reveal animations

localStorage API

Theme, color & visitor name persistence

Canvas API

Particle background with mouse repulsion

Google Fonts

DM Serif Display, DM Mono, DM Sans

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AI Integration

Claude by Anthropic

Code Generation

Scaffolded new features (sort, timer, visitor name, skill bars, timeline) matching the existing code style and CSS variable system.

Debugging

Identified a visitor greeting display bug where the hidden class wasn't removed before setting display values.

Architecture

Suggested re-appending DOM nodes for sorting instead of rebuilding the grid — more performant and preserves hidden card state.

Documentation

Assisted with writing ai-usage-report.md, technical-documentation.md, and the comprehensive README across all assignments.

All AI-generated code was reviewed, tested, and modified. I understand every line of this submission.

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Challenges & Lessons Learned

✅ Resolved

- GitHub Pages permissions error — solved by moving project to Desktop before pushing
- Visitor greeting not appearing — fixed by removing hidden class before changing display
- Sort function rebuilding entire card list — refactored to re-append only visible nodes
- CSS variable conflicts between A2 and A3 additions — resolved by namespacing new classes

🚀 Planned Future Work

- Backend integration for real contact form submissions
- Blog section with markdown-rendered posts
- Automated deployment pipeline with GitHub Actions
- Custom domain (abdullah-alzahrani.com)

4	18+	1	100%
Assignments Built	Features Implemented	Live Deployment	Vanilla JS — No Frameworks

Thank You

for watching

🌐 https://abdullah-zh.github.io/s202265440-Abdullah-Alzahrani-Assignment4/

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